Question 1

Revisit Later

You can write your answers in the provided space or write your answer on a piece of paper and scan and upload the handwritten answers using the QR Code available in the Scan and Upload Section of this exam.

Kindly ensure all answer sheets to be uploaded against the corresponding questions only. All sheets should not be uploaded against one question.

a) A telecom company is assessing the performance of its customer support center using the following customer waiting times (in minutes):

2.1, 3.5, 4.0, 5.2, 6.8, 2.9, 3.0, 4.5, 7.2, 8.0, 1.5, 2.8, 3.7, 4.9, 6.0, 7.5, 2.2, 3.3, 4.1, 5.8

(i) Calculate the five-number summary for the data. **[2 Marks]**

(ii) Interpret the results and discuss what they indicate about: **[1 Mark]**
- a) Customer service efficiency.
- b) The presence of unusually long waiting times.

b) In a e-coding competition, participant A has a 50% chance of completing the final challenge. If A fails to complete the challenge, participant B wins with probability 0.85. If A completes the challenge, participant B wins with probability 0.25. Determine the probability that participant B wins the competition. **[2 Marks]**

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Question 2

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The National Testing Agency (NTA) conducted a post-examination analysis of 600 NEET 2025 aspirants to study subject-wise performance in Physics (P), Chemistry (C), and Biology (B). The following information was obtained:

$$|P| = 380$$
$$|C| = 260$$
$$|B| = 420$$
$$|P \cap C| = 150$$
$$|P \cap B| = 250$$
$$|C \cap B| = 180$$

Assume that every student qualified in at least one subject.
Answer the following:

a) Determine the number and probability of students who qualified in all three subjects. **[1 Mark]**

b) Determine the probability that a randomly selected student qualified in exactly two subjects. **[1 Mark]**

c) Determine the probability that a student qualified in Physics but not Biology. **[1 Mark]**

d) Determine the probability that a student qualified in only Biology. **[1 Mark]**

e) Determine the probability that a student qualified in Physics or Biology but not Chemistry. **[1 Mark]**

f) Based on your calculations, comment on the overall overlap among the three subjects and its implications for student performance patterns. **[1 Mark]**

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Question 3

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A manufacturer of AI accelerator chips reports that the probability of a chip being defective is 0.03. A quality control engineer randomly inspects 40 chips from a large shipment.

i) Find the probability that exactly 4 chips are defective. **[1 Mark]**

ii) Find the probability that between 2 and 5 chips (inclusive) are defective. **[2 Marks]**

iii) The shipment is flagged for further inspection if the sample contains more than 4 defective chips. Compute the probability that the shipment is flagged. **[1 Mark]**

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Question 4

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Attempted: 0/6

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The lifetime (in years) of a certain type of battery is modelled by the probability density function:

$$f(x) = \begin{cases} \frac{x}{50}, & 0 \le x \le 10 \\ 0, & \text{otherwise} \end{cases}$$

i. Is f(x) a valid probability density function?
ii. Find the probability that a battery lasts more than 8 years.
iii. Find the expected lifetime E(X).
iv. Find the variance of X.
v. Interpret the results in terms of product reliability.

[5 Marks]

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B I U x² x₂

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Attempted: 0/6

Question 5

Revisit Later

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A company wants to classify whether a customer will buy a product (Yes/No) based on two features: **Age Group** and **Income Level**. Past data is summarized below:

Age Group	Income Level	Buy Yes	Buy No
Young	Low	5	15
Young	High	20	5
Middle	Low	10	10
Middle	High	30	10
Senior	Low	3	12
Senior	High	12	8

A prospective customer belongs to the Middle age group and has a High income level.

Using the Naive Bayes assumption of conditional independence approach:

a) Compute the prior probabilities of the target classes. **[1 Mark]**

b) Calculate the likelihood probabilities associated with the customer's attributes. **[1 Mark]**

c) Determine the posterior probabilities for both classes (Buy = Yes and Buy = No) **[2 Marks]**

d) Predict whether the customer is likely to purchase the product and interpret the result from a business perspective. **[1 Mark]**

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a) A bank uses an automated system to flag potentially fraudulent transactions. From historical data, 2% of all transactions are fraudulent. If a transaction is fraudulent, the system correctly flags it 90% of the time. If a transaction is not fraudulent, the system incorrectly flags it as suspicious 5% of the time. One day, a customer's transaction is flagged as suspicious. **[3 Marks]**

(i) Compute the probability that the flagged transaction is actually fraudulent.

(ii) Comment on whether the system's flag strongly indicates fraud, and discuss implications for customer trust.

b) An e-commerce company records the delivery times (in days) for its packages. Data analysis shows that delivery times follow a normal distribution with a mean of 5 days and a standard deviation of 1.2 days. **[2 Marks]**

i. What is the probability that a package is delivered in less than 4 days.
ii. Find the probability that a package is delivered between 4 and 6 days.

z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-0	.50000	.49601	.49202	.48803	.48405	.48006	.47608	.47210	.46812	.46414
-0.1	.46017	.45620	.45224	.44828	.44433	.44034	.43640	.43251	.42858	.42465
-0.2	.42074	.41683	.41294	.40905	.40517	.40129	.39743	.39358	.38974	.38591
-0.3	.38209	.37828	.37448	.37070	.36693	.36317	.35942	.35569	.35197	.34827
-0.4	.34458	.34090	.33724	.33360	.32997	.32636	.32276	.31918	.31561	.31207
-0.5	.30854	.30503	.30153	.29806	.29460	.29116	.28774	.28434	.28096	.27760
-0.6	.27425	.27093	.26763	.26435	.26109	.25785	.25463	.25143	.24825	.24510
-0.7	.24196	.23885	.23576	.23270	.22965	.22663	.22363	.22065	.21770	.21476
-0.8	.21186	.20897	.20611	.20327	.20045	.19766	.19489	.19215	.18943	.18673
-0.9	.18406	.18141	.17879	.17619	.17361	.17106	.16853	.16602	.16354	.16109
-1	.15866	.15625	.15386	.15151	.14917	.14686	.14457	.14231	.14007	.13786
-1.1	.13567	.13350	.13136	.12924	.12714	.12507	.12302	.12100	.11900	.11702
-1.2	.11507	.11314	.11123	.10935	.10749	.10565	.10383	.10204	.10027	.09853
-1.3	.09680	.09510	.09342	.09176	.09012	.08851	.08692	.08534	.08379	.08226
-1.4	.08076	.07927	.07780	.07636	.07493	.07353	.07215	.07078	.06944	.06811
-1.5	.06681	.06552	.06426	.06301	.06178	.06057	.05938	.05821	.05705	.05592
-1.6	.05480	.05370	.05262	.05155	.05050	.04947	.04846	.04746	.04648	.04551
-1.7	.04457	.04363	.04272	.04182	.04093	.04006	.03920	.03836	.03754	.03673
-1.8	.03593	.03515	.03438	.03362	.03288	.03216	.03144	.03074	.03005	.02938
-1.9	.02872	.02807	.02743	.02680	.02619	.02559	.02500	.02442	.02385	.02330
-2	.02275	.02222	.02169	.02118	.02068	.02018	.01970	.01923	.01876	.01831

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